NATIONAL UNIVERSITY OF TECHNOLOGY ISLAMABAD

DEPARTMENT OF COMPUTER SCIENCE

DIGITAL LOGIC DESIGN LAB

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Topic Covered	**Design and Analysis of a 1-Bit Binary Comparator**

Experiment 12

Design and Analysis of a 1-Bit Binary Comparator

(Open Ended Lab)

Objective

- To understand the working of a 1-bit binary comparator.
- To design logic expressions and circuits that compare two single-bit binary inputs.
- To learn the role of XNOR, AND, and NOT gates in implementing comparison logic.

Design Problem

Design a combinational logic circuit that compares two 1-bit binary inputs, A and B. The circuit should generate three outputs:

$$L = 1 \; if \; A < B$$
$$E = 1 \; if \; A = B$$
$$G = 1 \; if \; A > B$$

Tasks:

1. Write the truth table for all possible combinations of A and B.

TRUTH TABLE

A	B	L	E	G
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

2. Derive **simplified logic expressions** for each output

   From the truth table:

   - **L (A < B)** → `L = A' · B`
   - **E (A = B)** → `E = A XNOR B` → `E = A·B + A'·B'`

- **G (A > B)** → `G = A · B'`

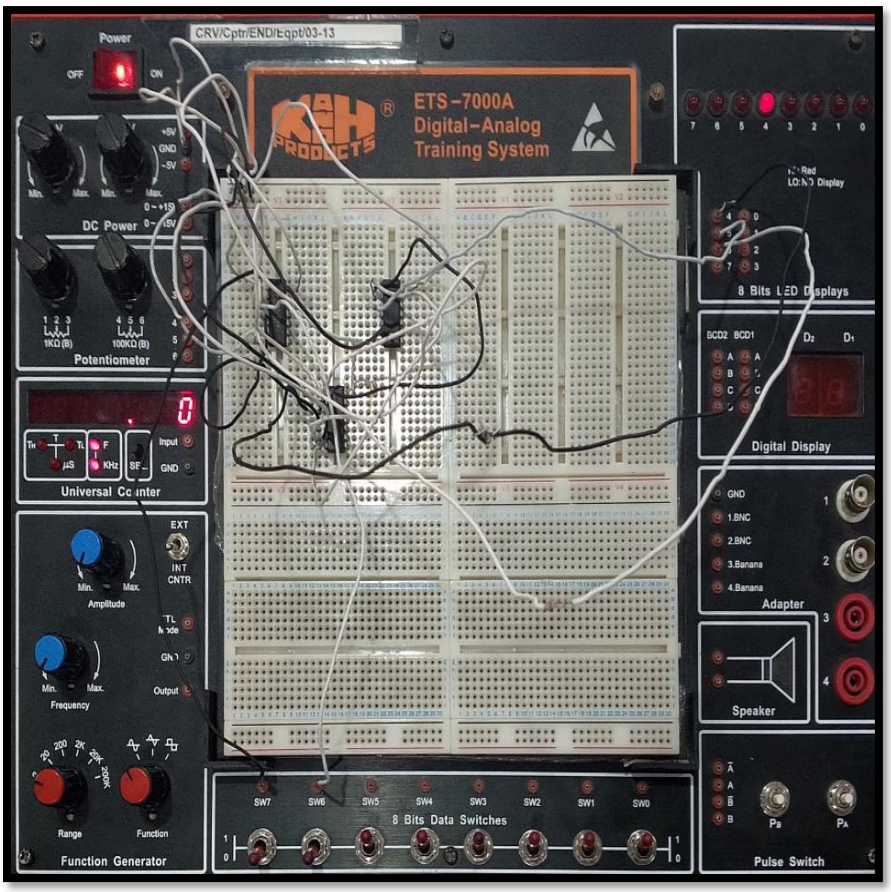
3. Implement the circuit using basic logic gates and verify the output using a logic trainer kit.

Basic Gate Implementation:

- **NOT Gates**: For `A'` and `B'`
- **AND Gates**:
  - `A' · B` → Output **L**
  - `A · B'` → Output **G**
  - `A · B` and `A' · B'` → fed into OR gate to get **E**

- **XNOR Gate** (Optional): Directly get **E** if available

CIRCUIT DIAGRAM



:. IS THIS FIGURE WE JUST SHOW THE OUTPUT WHRER A=B

4. Analyze how the gates contribute to equality detection in the comparator design.

Logic Gates:

- **XNOR Gate:** Produces an output when both inputs match (both 0 or both 1).
- **AND Gate:** Generates an output only when specific input conditions are satisfied (e.g., A and B meet certain criteria).
- **NOT Gate**: Inverts the input signal, flipping its logical state (0 becomes 1, and 1 becomes 0).

Tools/Equipment Requirement

Required Equipment:

- Logic Trainer Kit
- Connecting Wires
- Logic Gate ICs:
    - 7404 (NOT)
    - 7408 (AND)
    - 7432 (OR)
    - 7486 (XOR) or 74266 (XNOR)
- 5V DC Power Supply

Learning outcomes:

- **Binary Comparison:** Learned to compare two binary inputs and determine their relationship (equal, greater than, or less than).
- **Comparator Circuit Design:** Understood how to design a basic comparator circuit using logic gates (AND, OR, NOT).
- **Digital Decision-Making:** Gained insight into how digital systems make decisions based on binary values.
- **Practical Applications:** Recognized the importance of comparators in processors, sorting circuits, and control systems.